

Mapping Mental Health and Social Stressors on Reddit: A Text Classification Pipeline

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Abstract

Social media platforms have become important venues for people to disclose personal struggles that intersect with mental health and major life stressors. This project develops an end-to-end machine-learning pipeline that classifies Reddit posts into six categories related to mental health and social stressors: addiction, alcoholism, attention-deficit/hyperactivity disorder (ADHD), bipolar disorder, depression, and divorce. The work is motivated by the need for scalable, population-level insight into patterns of distress that are global in scope, but highly relevant to Canada and, in particular, to the Atlantic provinces.

Introduction

Over the past decade, social media has become part of our everyday lives, with online platforms now hosting important discussions about mental health, well-being, and individual social life experiences. Because of the level of anonymity provided by platforms such as Reddit and its communities, it has become a natural place for people to discuss their personal experiences. Posts in some of these communities often describe severe life circumstances.

Mental-health and social-stressor conversations are now a global challenge. Across countries, people use platforms such as Reddit to disclose deeply personal struggles with addiction, alcoholism, ADHD, bipolar disorder, depression, and divorce. These posts are often written at crisis points: when someone is relapsing into substance use, overwhelmed by untreated ADHD, destabilized by bipolar mood swings, trapped in a depressive episode, or navigating the emotional and practical chaos of marital breakdown. Although these conversations contain rich signals about people's men-

tal health and support needs, they appear as an unstructured stream of millions of posts distributed across different communities. Human moderators, clinicians, and public health practitioners cannot manually read or monitor this volume of content in real time.

Epidemiological data and clinical observations indicate that mental-health disorders and associated social stressors constitute a leading cause of disability worldwide. Canada, despite its relatively high-income status and publicly funded health system, faces significant mental-health burdens, with uneven access to specialized care across regions. Within Canada, the Atlantic provinces: Nova Scotia, New Brunswick, Prince Edward Island, and Newfoundland and Labrador, experience their own mix of economic vulnerability, rural and remote disparities, and limited access to mental-health services. The result is that many individuals in distress will not appear in formal health-system data, either because they do not seek care or because they experience delays. In contrast, these individuals may be present in online spaces. Social media data represents a rich source of information for evaluating common and large-scale experiences of individuals in distress and in need of support, which traditional healthcare data might miss.

This work frames the challenge as a supervised, multi-class text classification problem. Given a Reddit post (title and body), the model automatically predicts which of six social-mental health categories it most closely reflects: addiction, alcoholism, ADHD, bipolar disorder, depression, or divorce. Each category captures a distinct type of psychosocial stressor and risk profile: addiction and alcoholism relate to substance dependence and relapse risk; ADHD to chronic difficulties with attention, impulse control, and functioning; bipolar disorder to severe mood instability; depression to persistent low mood and suicidality risk; and divorce to acute life-change stress, loss, and family disruption. Importantly, divorce itself is not a mental-health diagnosis but a major social stressor; it is included deliberately as a contrasting class to help the model distinguish between posts about psychiatric conditions and posts about life events that can trigger or exacerbate those conditions. By training on labelled posts from these topic-specific subreddits, the system learns how people organically describe these problems in their own words, and can then map new, unseen posts to one of these categories at scale.

We used Reddit Mental Health Dataset [1], which is publicly available via Zenodo (<https://zenodo.org/records/3941387>). This dataset contains several CSV files per subreddit and time period. For this project, six “pre” files are used, corresponding to the addiction, alcoholism, ADHD, bipolar, depression, and divorce subreddits. These files are placed into a dedicated `data/` directory at the project root. Although the original files contain precomputed TF-IDF features, we re-parsed the raw text components in order to construct create a reproducible feature pipeline. The data contains four columns in each CSV, and we interpret them as subreddit name, author, date, and post content. We ensured to use a balanced dataset across all categories to prevent any high-activity subreddit community from dominating our model’s loss function. Each of the six classes contributes 2613 posts to overall datasets as shown in Figure 1 and the most frequent words used are shown in Figure 2.

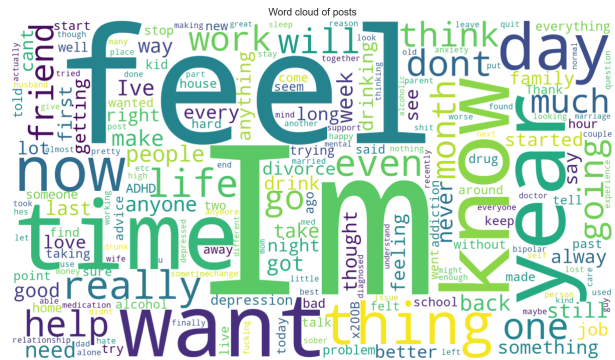


Figure 2: Word cloud visualizing the most frequent terms across analyzed subreddits, with prominent themes including personal experience ("feel," "I'm," "know") and temporal markers ("time," "year," "day").

As part of our data processing phase, we normalized column names to `Subreddit`, `Date`, `Title`, and `Snippet`. We concatenated `Title` and `Snippet` into a single `content` field, removed non-ASCII characters and HTML artifacts, converted date strings into `datetime` objects, and filtered the dataset to only posts before a cutoff to reduce random and temporal heterogeneity. We used

the Natural Language Toolkit (NLTK) to implement several standard steps: tokenization using the Punkt sentence and word tokenizers; stopwords removal based on the English stopwords list, augmented with high-frequency conversational tokens such as “im” and “hey”; and lemmatization using WordNet to reduce inflected forms to canonical lemmas. We also converted text to lowercase and stripped of punctuation before tokenization. The preprocessed tokens were rejoined into a cleaned string, which served as input to the feature extractor.

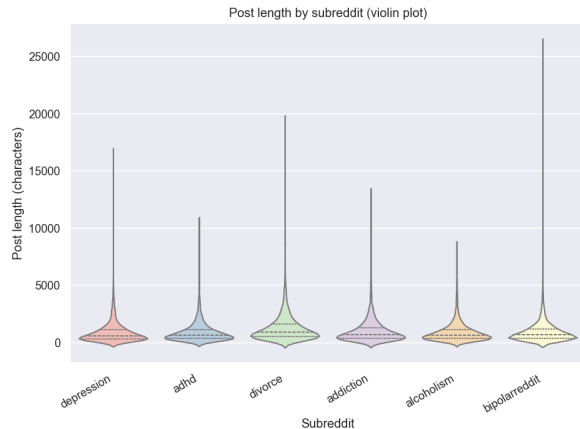


Figure 3: Violin plot showing character length distribution across specific subreddits.

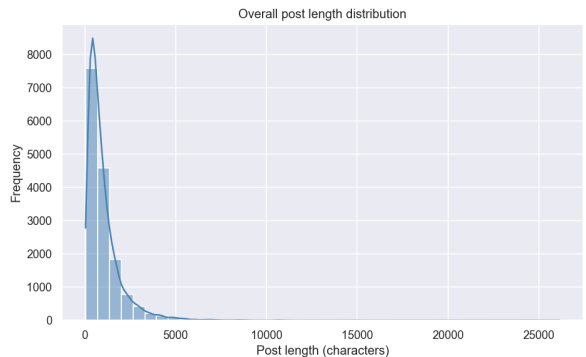


Figure 4: Aggregated distribution of post lengths across the entire dataset.

The scikit-learn package `TfidfVectorizer` was used for feature extraction. This transformer package maps clean text into a high-dimensional sparse vector space whose components correspond to unigrams and bigrams, with a maximum of 500 features retained. The minimum document frequency parameter we used was zero in order to retain terms that appear in any fraction of documents. The maximum document frequency was set to one to prevent common terms from dominating. Figures 3 and 4 illustrate the overall post length distributions, respectively, providing a comprehensive overview of the data.

Models training and evaluation

Model training and evaluation are encapsulated in `model.py` and `evaluate.py`. The labelled dataset was split into training and test partitions using an 80/20 ratio with a fixed random seed after the TF-ID transformation on the feature matrices and corresponding label vectors. Three candidate model classifiers were considered: Logistic Regression using a one-vs-rest scheme, Gra-

dient Boosting, and Random Forests. The Logistic Regression model used the `lbfgs` solver with an increased maximum number of iterations (300) to ensure convergence on the training data. Gradient Boosting and Random Forest classifiers were configured with 100 estimators each, with moderate depths to balance bias and variance. To exclude cross-validation during model training, we set a `--no-cv` flag to the main runner script; here, the models simply train once on the training set and are evaluated on the held-out test set. In our configuration, we compute the weighted F1-scores for each model on the test set and select the model with the highest F1-score as the “best” classifier.

We evaluated our models by calculating standard classification metrics, including per-class precision, recall, and F1-score, and overall accuracy. For Logistic Regression, the per-class performance is as follows: addiction shows a precision of 0.82, recall of 0.78, and F1-score of 0.80 (546 test examples); ADHD yields 0.77 precision, 0.78 recall, and 0.78 F1-score (485 examples); alcoholism achieves 0.86 precision, 0.82 recall, and 0.84 F1-score (535 examples); bipolarreddit attains 0.77 precision, 0.71 recall, and 0.74 F1-score (527 examples); depression exhibits 0.71 precision, 0.82 recall, and 0.76 F1-score (503 examples); and divorce achieves the highest scores with 0.91 precision, 0.92 recall, and 0.91 F1-score (540 examples). The macro and weighted averages of the F1-score both approximate 0.81, indicating broadly consistent performance across classes, albeit with some confusion between the depression and bipolarreddit categories. We used the confusion matrix to make misclassification patterns more apparent for the models. For example, the Logistic Regression shows that misclassifications tend to occur between depression and bipolar reddit rather than between, say, alcoholism and divorce, which is intuitive given the semantic overlap between depressive and bipolar narratives.

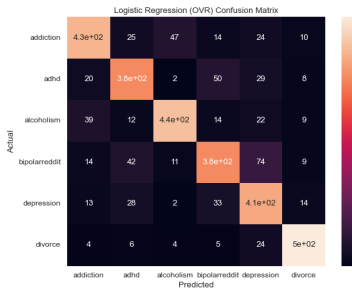


Figure 5: Confusion matrix for the Logistic Regression model.

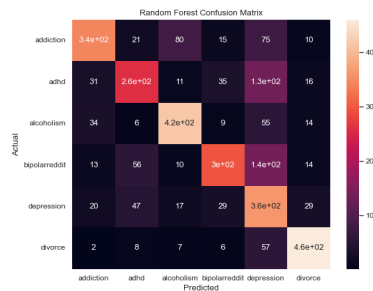


Figure 6: Confusion matrix for the Random Forest model.

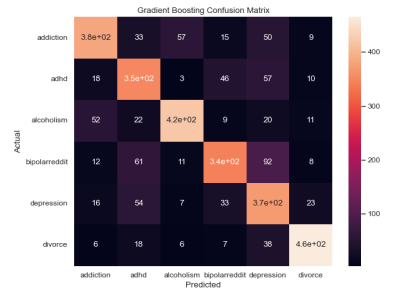


Figure 7: Confusion matrix for the Gradient Boosting model.

We finally used the multiclass ROC curves using a one-vs-rest formulation. For each class, the true-positive rate is plotted against the false-positive rate across thresholds, and an area under the curve (AUC) was computed. A micro-average ROC curve aggregates all classes to provide an overall sense of discriminative performance.

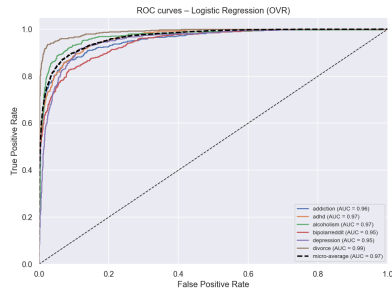


Figure 8: ROC curves for Logistic Regression classification.

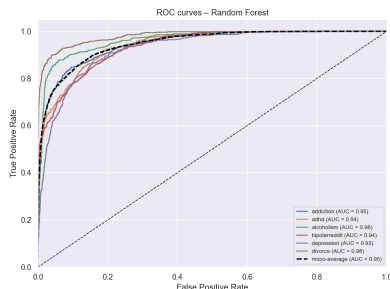


Figure 9: ROC curves for Random Forest classification.

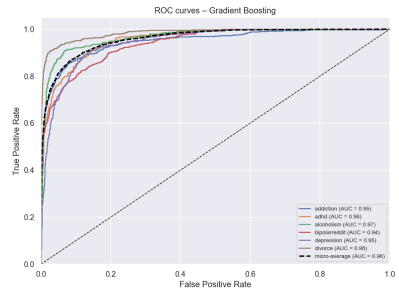


Figure 10: ROC curves for Gradient Boosting classification.

Rationale for tool selection and alternatives considered

Our choice of models and tools reflects a deliberate preference for simplicity and interpretability. Python, together with pandas and NumPy, provides a familiar environment for data manipulation. NLTK, while not the most modern NLP library, offers robust tokenization and lemmatization and is more than adequate for pre-processing Reddit posts. Scikit-learn remains a standard for TF-IDF, classical classifiers, and metrics, and its API encourages clear separation between preprocessing, model fitting, and evaluation. We used matplotlib and seaborn libraries for our visualizations. Transformer-based language models, such as BERT, often outperform bag-of-words methods on nuanced classification tasks and can capture context and subtle semantics more effectively. However, they demand greater computational resources and introduce additional complexity in fine-tuning and interpretability. Given that the primary aim of this project is to demonstrate a complete and comprehensible pipeline that can be reproduced easily and communicated to stakeholders, classic TF-IDF with Logistic Regression, Gradient Boosting, and Random Forests represents a reasonable and principled choice.

Similarly, in reality, posts in mental-health subreddits may likely belong to more than one category, such as an individual discussing both alcohol use and depressive symptoms. A multi-label classifier would likely provide a more representative view of this reality. Nonetheless, the

labels in the Reddit Mental Health Dataset are inherently single-label, attached to the subreddit rather than to manually annotate themes. For these reasons, a multi-class formulation was adopted here, with the understanding that it approximates but does not fully capture the complexity of the real world.

Reflections

Despite the simplicity of our approach, performance was respectable, with an overall F1-score of approximately 0.81 and generally strong precision and recall for most classes. This reinforces a lesson often emphasised in the literature but sometimes overlooked in practice: for many text classification problems, especially those with moderately sized vocabularies and clear class boundaries, linear models on top of bag-of-words features remain highly competitive. More so, the alignment of our project with public health in Atlantic Canada also proved fruitful. Even though our dataset itself is not region-specific, it is possible to frame our results in terms of the types of issues, such as substance use, mood instability, and family disruption, that Atlantic Canadian communities recognize as pressing. This connection made this work feel less abstract, and similar analyses could inform local service planning if region-specific data is available.

Challenges and limitations

We encountered a number of challenges during the project. The most fundamental is the inherent noise in subreddit labels. Membership in a subreddit is an imperfect proxy for a person’s mental-health status or for the primary theme of a post. Individuals may post in the “wrong” community, may cross-post similar content to multiple subreddits, or may use a community as a general outlet rather than for its nominal topic. As a result, the classifier is, at best, learning to recognize patterns that correlate with community membership, not formal diagnoses.

Another challenge is the overlap between certain categories. The confusion matrix revealed that misclassifications were particularly common between depression and bipolar disorder, which is unsurprising given that bipolar disorder includes depressive episodes and that the language used to describe them can be very similar. This suggests that, in the absence of explicit temporal information about mood cycles, text-only models may struggle to discriminate reliably between

these two classes.

Future improvement

One obvious direction for improvement is to incorporate contextual embeddings, such as those provided by pre-trained transformer models, while still retaining the classical classifier layer. This would allow the model to exploit richer representations of meaning without fully committing to the complexity of end-to-end fine-tuning. A second direction would be to revisit the label structure. With additional annotation effort, it would be possible to construct a multi-label dataset in which posts are tagged with all relevant themes. A third improvement would focus on evaluation and deployment. Implementing a lightweight dashboard, perhaps using an interactive framework, would make it easier for public health practitioners or community organizations to explore class distributions over time, examine key phrases associated with each category, and experiment with hypothetical posts. Incorporating fairness and bias analyses, for instance, by examining whether misclassification rates differ systematically across imagined demographic profiles within the text, would also be a valuable extension.

Conclusion

In summary, we have demonstrated that a relatively modest set of tools, carefully assembled and critically interpreted, can yield a useful and extensible system for mapping mental-health-related discourse on social media. It also shows that, whilst computational methods cannot substitute for clinical judgement or detailed local knowledge, they can offer a valuable additional lens through which to view the evolving landscape of mental health and social stressors.

References

- [1] Low, D. M., Rumker, L., Torous, J., Cecchi, G., Ghosh, S. S., & Talkar, T. (2020). Natural Language Processing Reveals Vulnerable Mental Health Support Groups and Heightened Health Anxiety on Reddit During COVID-19: Observational Study. *Journal of Medical Internet Research*, 22(10), e22635.